

TrES-5b

R-0952

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_230627 · created 2026-07-29T14:42:22+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460122.699 +/- 0.0037 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.174 +/- 0.048
Transit depth (Rp/Rs)^2	3.02 +/- 1.65 [%]
Orbital Inclination (inc)	81.75 +/- 2.78
Airmass coefficient 1 (a1)	1.188 +/- 0.038
Airmass coefficient 2 (a2)	-0.132 +/- 0.023
Scatter in the residuals of the lightcurve fit is	3.11 %
Best Comparison Star	#3 - [222, 453]
Optimal Method	PSF photometry
Transit Duration (day)	0.049 +/- 0.028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.174 ± 0.048 vs 0.142 (deviation 0.67σ)
Duration fit vs published	0.049 d vs 0.0758 d (deviation 0.96σ)
Mid-time O–C	-9.7 min
Depth SNR	3.6
Residual scatter	3.11 %

Light curve

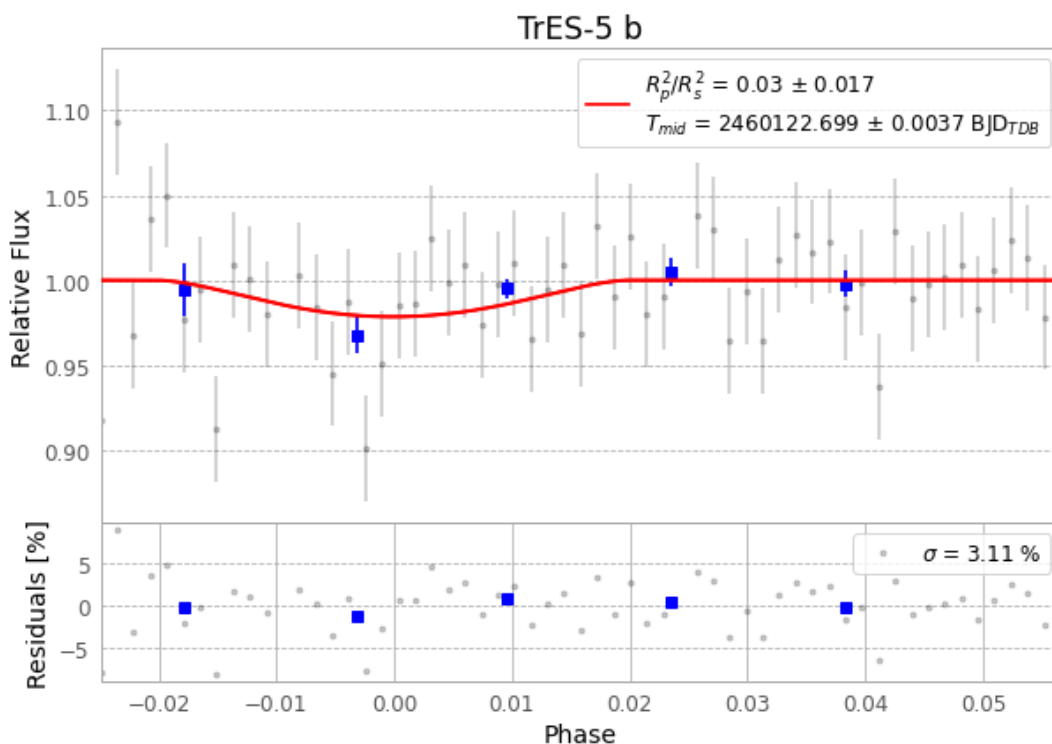


Figure 1 — FinalLightCurve_TrES-5 b_2023-06-26.png (SHA-256 edb1c8963428c709...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [271, 241], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 30b94862350c6f73...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

59 FITS frames (39 MB), TRES-5230627035115.FITS → TRES-5230627064516.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-230627.log (2a55c75c4463...), FinalLightCurve_TrES-5 b_2023-06-26.png (edb1c8963428...), FinalLightCurve_TrES-5 b_2023-06-26.csv (4709e060b509...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 14:42Z.